Lecture 18: The Exponential Function

MATH203 Applied Linear Algebra

What Is e?

A Question About Money

You deposit \$1 in a bank. The bank's rule: **100% interest per year**, computed **linearly**.

- After 1 year: $\$1 + 100\% = \$2$
- After 2 years: $\$1 + 200\% = \$3$
- After half a year: $\$1 + 50\% = \$1.50$

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Better trick: Withdraw after **half** a year (\$1.50), re-deposit.

$$\text{After 1 year} = \$1.50 + 50\% \times \$1.50 = \$2.25 \quad \text{(instead of \$2!)}$$

The Greedy Strategy

Idea: Withdraw and re-deposit as **frequently as possible**.

Every $\frac{1}{n}$ of a year, take out your money and put it back:

$$\text{After each period:} \quad \text{balance} \times \left(1 + \frac{1}{n}\right)$$

After n periods (one full year):

$$\text{Final balance} = \$1 \times \left(1 + \frac{1}{n}\right)^n$$

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n (withdrawals per year)	$\left(1 + \frac{1}{n}\right)^n$
1	2.000
2	2.250
4	2.441
10	2.594
100	2.705
1000	2.717
∞	$2.71828\ldots$

The Number e

Definition

$$e := \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = 2.71828\ldots$$

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More generally, if the interest rate is x (instead of $100\% = 1$):

The Exponential Function

$$e^x := \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

This is our **definition** of the exponential function. Everything else follows from it.

From Limit to Power Series

The Binomial Theorem

Recall the binomial expansion for any positive integer n:

$$(1+x)^n = 1 + \frac{n}{1!}\,x + \frac{n(n-1)}{2!}\,x^2 + \frac{n(n-1)(n-2)}{3!}\,x^3 + \cdots$$

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Now think of e^x as $\left(1+\frac{x}{\infty}\right)^{\infty}$.

Apply the binomial expansion with $x \mapsto \frac{x}{n}$ and $n \to \infty$:

$$\left(1+\frac{x}{n}\right)^n = 1 + \frac{n}{1!}\cdot\frac{x}{n} + \frac{n(n-1)}{2!}\cdot\frac{x^2}{n^2} + \frac{n(n-1)(n-2)}{3!}\cdot\frac{x^3}{n^3} + \cdots$$

The Coefficients Collapse

Look at each coefficient as $n \to \infty$:

$$\frac{n}{n} = 1, \qquad \frac{n-1}{n} \to 1, \qquad \frac{n-2}{n} \to 1, \qquad \ldots$$

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The Exponential Series

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots}$$

This is not a separate formula — it is the **binomial expansion** of the limit definition.

Why This Matters

The power series $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ gives us:

1. A way to **compute** e^x for any x (plug in and add terms)
2. A way to **substitute anything** for x — a complex number, a matrix, a derivative operator
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The philosophy:

Limit definition tells you **what** e^x *is.*

Power series tells you **how to work with it**.

Exponential Solves $y' = ky$

A Question About Motion

Differential equation: $y' = 2y$.

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Question: Where are you at time s?

Strategy: Same as compound interest — break time into tiny steps.

Tiny Time Steps

Break the interval $[0, 1]$ into n steps of size $\frac{1}{n}$.

At time 0: position $= y(0)$, speed $= 2y(0)$.

After one tiny step:

$$y\left(\frac{1}{n}\right) \approx y(0) + \frac{1}{n} \cdot 2y(0) = y(0)\left(1 + \frac{2}{n}\right)$$

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After two steps:

$$y\left(\frac{2}{n}\right) \approx y\left(\frac{1}{n}\right)\left(1 + \frac{2}{n}\right) = y(0)\left(1 + \frac{2}{n}\right)^2$$

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After n steps (time $= 1$):

$$y(1) = y(0)\left(1 + \frac{2}{n}\right)^n$$

Taking the Limit

As $n \to \infty$ (steps become infinitely small):

$$y(1) = \lim_{n\to\infty} y(0)\left(1 + \frac{2}{n}\right)^n = y(0) \cdot e^2$$

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For general time s: need ns steps to reach time s, so

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Solution

The differential equation $y' = 2y, \quad y(0) = 1$ has solution:

$$\boxed{y(s) = e^{2s}}$$

The General Pattern

Theorem

The differential equation $y' = ky, \quad y(0) = c$ has solution:

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Why? Same argument:

$$y(s) = y(0)\left(1 + \frac{k}{n}\right)^{ns} \longrightarrow y(0) \cdot e^{ks}$$

The exponential function is what you get when the rate of change is proportional to the current value.

$k > 0$	Exponential **growth**
$k < 0$	Exponential **decay**
$k = 0$	Constant

The Same Idea, Twice

Compound Interest	Differential Equation
Rate $= x$	Rate $= k$
n compounding periods	n time steps
Each period: multiply by $\left(1 + \frac{x}{n}\right)$	Each step: multiply by $\left(1 + \frac{k}{n}\right)$
After 1 year: $\left(1 + \frac{x}{n}\right)^n$	After time 1: $\left(1 + \frac{k}{n}\right)^n$
$n \to \infty$: $\quad e^x$	$n \to \infty$: $\quad e^k$

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$n \to \infty$: $\quad e^x$	$n \to \infty$: $\quad e^k$

They are the same phenomenon.

The exponential function arises whenever:

"Repeated tiny multiplicative nudges compound into a smooth transformation."

Complex Exponential & Euler's Formula

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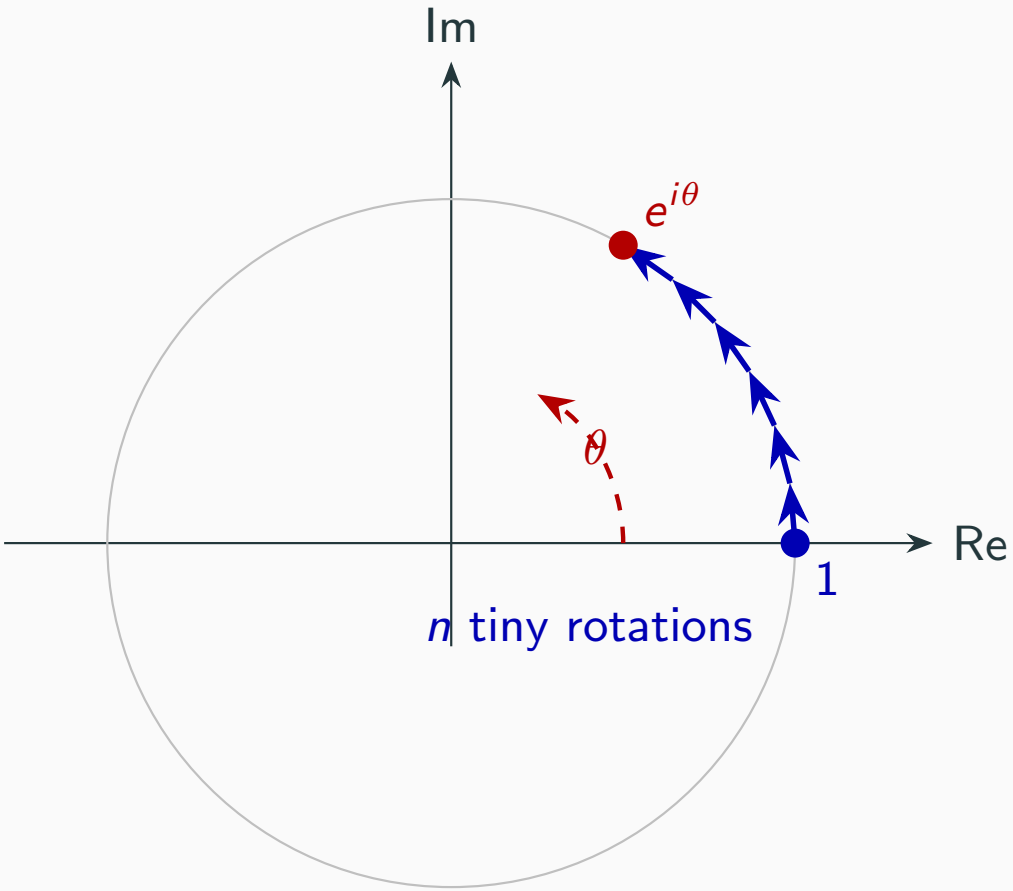
Each factor $\left(1 + \frac{i\theta}{n}\right)$ is a complex number with:

- Modulus ≈ 1 (no stretching)
- Argument $\approx \frac{\theta}{n}$ (tiny rotation)

So each multiplication is a **tiny rotation by θ/n**.

n Tiny Rotations = One Big Rotation



Multiply n factors of $\left(1 + \frac{i\theta}{n}\right)$:

- Each rotates by $\approx \theta/n$
- Total rotation $= n \times \frac{\theta}{n} = \theta$
- Total stretching ≈ 1 (modulus stays ≈ 1)

Geometric Proof: Why It Must Be a Circle

Consider $y(t) = e^{it}$ as a moving particle. Its position at time 0 is $y(0) = 1$.

The differential equation $y' = iy$ tells us:

- **Velocity** $= i\times$ position
- Multiplying by i = rotating by 90° (from L17)

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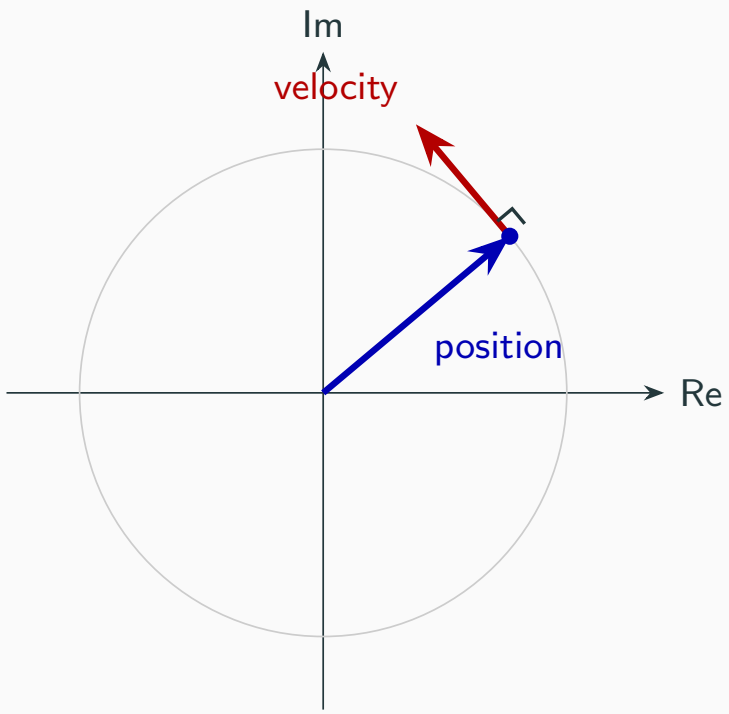
So the velocity is always **perpendicular** to the position vector!



The Circle Argument

Velocity perpendicular to position means:

1. The **distance from origin never changes**: if you're always moving sideways, you never get closer or farther. So $|e^{it}| = |e^{i \cdot 0}| = 1$ for all t.

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Euler's Formula (Geometric Proof)

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

$e^{i\theta}$ is the point on the unit circle at angle θ. That's it.

Euler's Formula from the Power Series (Second Proof)

For those who know the Taylor series of sin and cos:

Substitute $x = i\theta$ into $e^x = 1 + x + \frac{x^2}{2!} + \cdots$:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots$$

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Use $i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad \ldots$:

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{\cos\theta} + i\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{\sin\theta}$$

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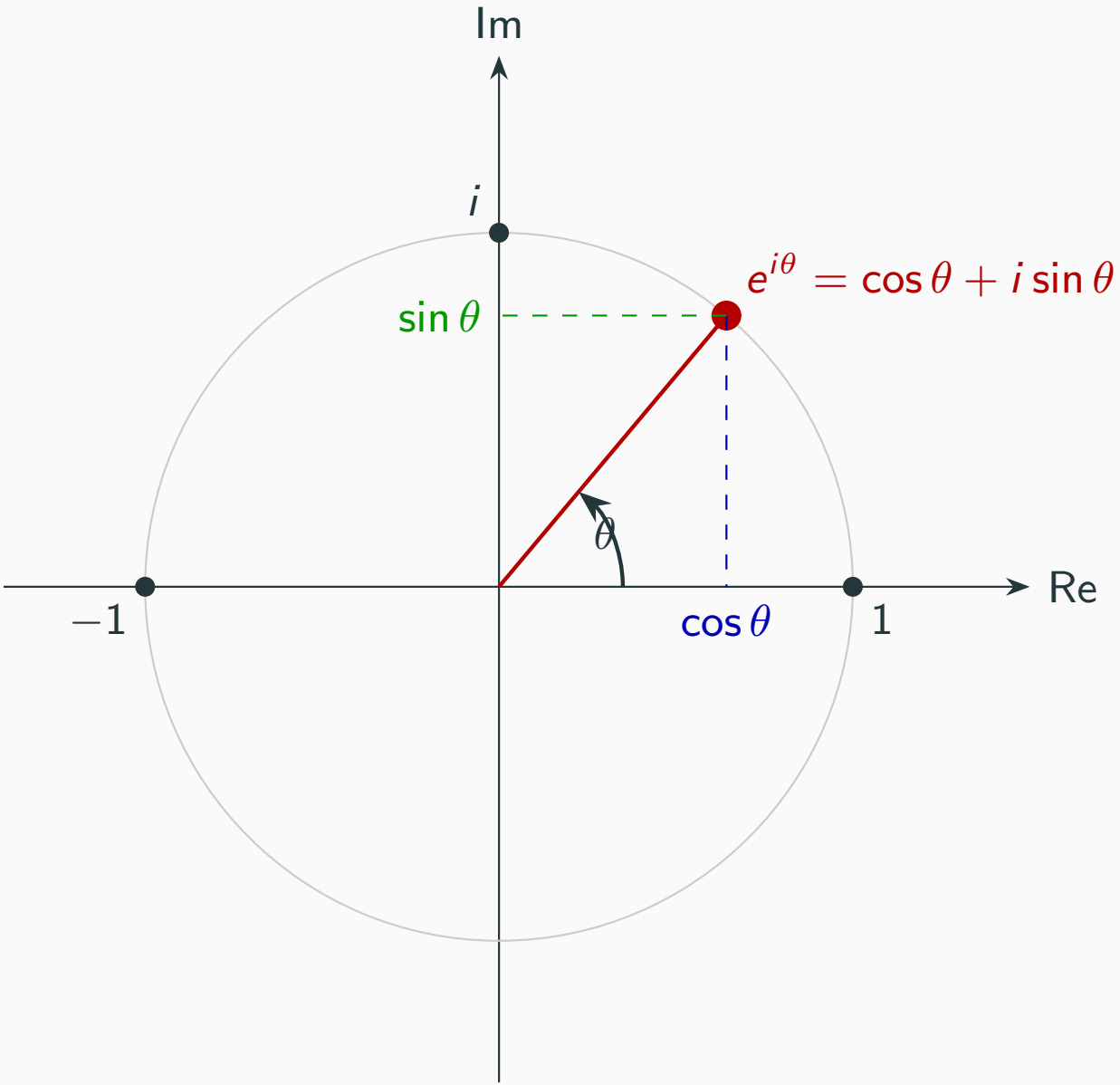
Same result. But note: this proof **assumes** you already know what $\cos\theta$ and $\sin\theta$ expand to. The geometric proof derives Euler's formula from first principles.

The Geometry of $e^{i\theta}$



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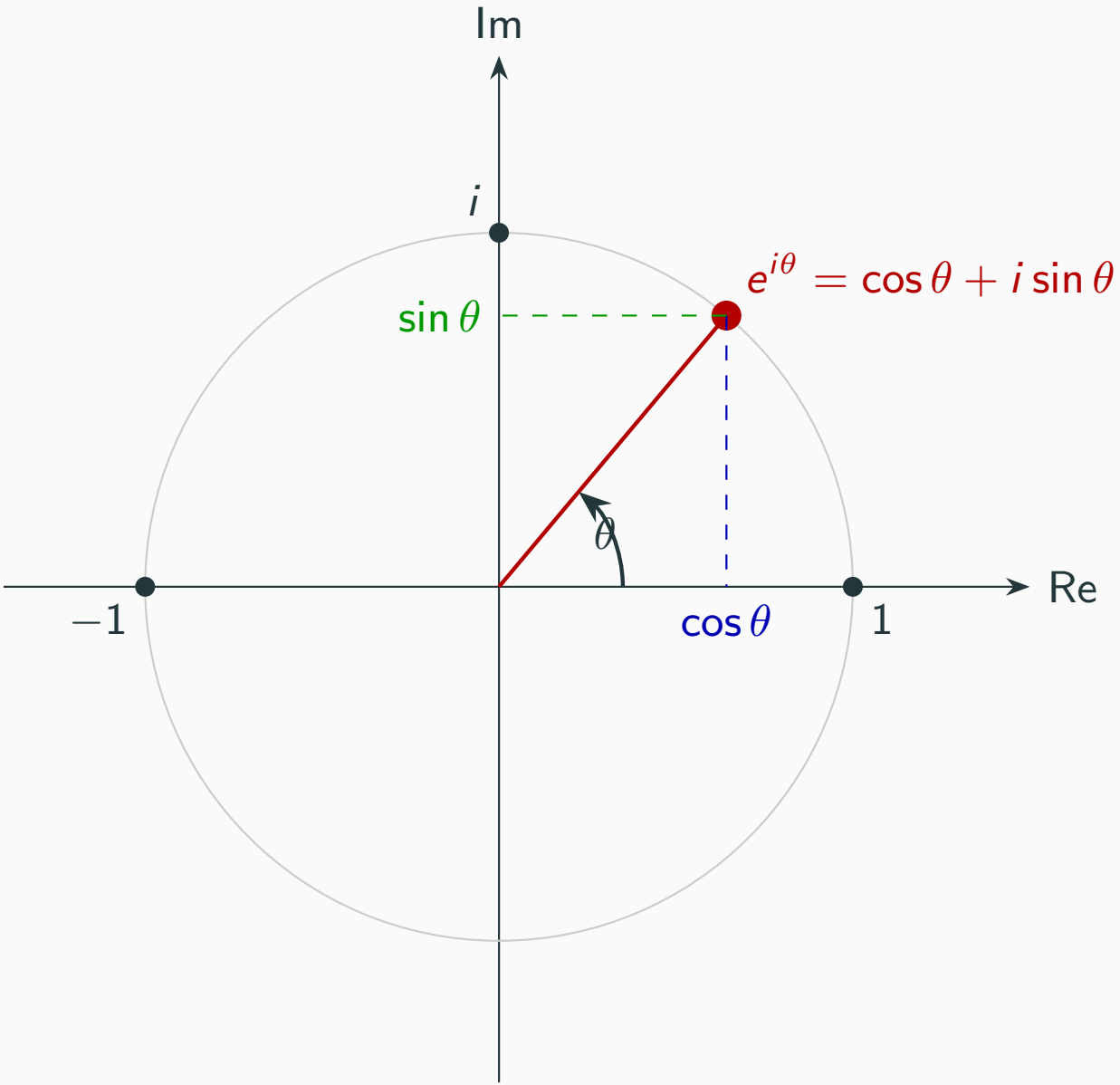

Exponential of $\frac{d}{dx}$ and Taylor's Theorem

What If We Exponentiate a Derivative?

We defined $e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$ for numbers.

Bold idea: Replace x with the operator $\frac{d}{dx}$:

$$\exp\left(\frac{d}{dx}\right) := \lim_{n\to\infty}\left(I + \frac{1}{n}\frac{d}{dx}\right)^n$$

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$$\exp\left(\frac{d}{dx}\right) := \lim_{n\to\infty}\left(I + \frac{1}{n}\frac{d}{dx}\right)^n$$

Question: What does this operator **do** to a function $f(x)$?

Strategy: understand what $\left(I + \frac{1}{n}\frac{d}{dx}\right)$ does, then apply it n times.

The Operator $I + \frac{1}{n}\frac{d}{dx}$

This operator maps:

$$f(x) \longmapsto f(x) + \frac{1}{n} f'(x)$$

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Try a linear function: $f(x) = kx + b$.

$$f(x) + \frac{1}{n}f'(x) = kx + b + \frac{k}{n} = k\left(x + \frac{1}{n}\right) + b = f\left(x + \frac{1}{n}\right)$$

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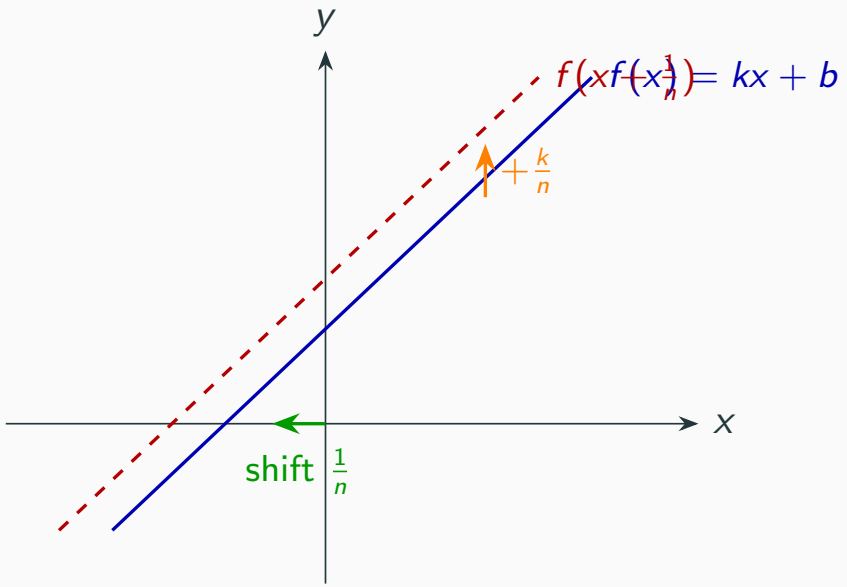
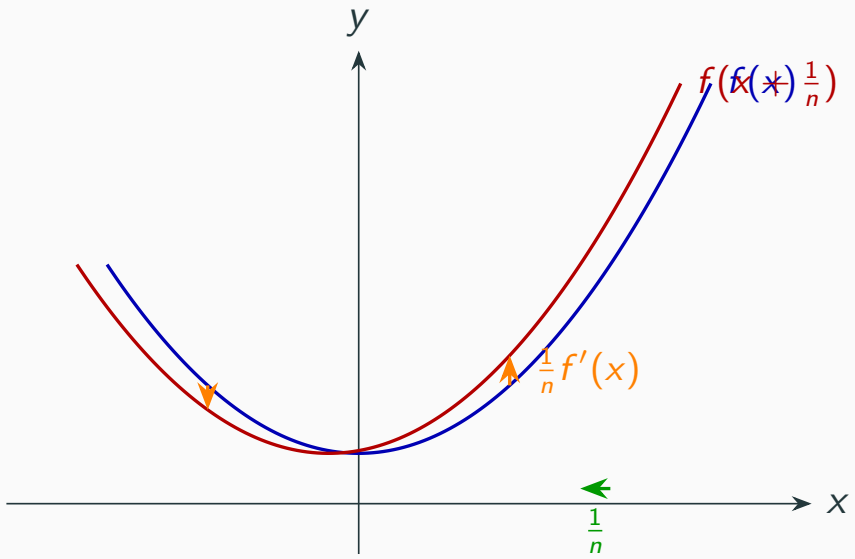


For a line: shifting **up** by $\frac{k}{n}$ = shifting **left** by $\frac{1}{n}$.

Any Smooth Function: Shift Up ≈ Shift Left

On a tiny interval, any smooth function is approximately linear. So the same logic applies:

$$f(x) + \frac{1}{n} f'(x) \;\approx\; f\left(x + \frac{1}{n}\right)$$



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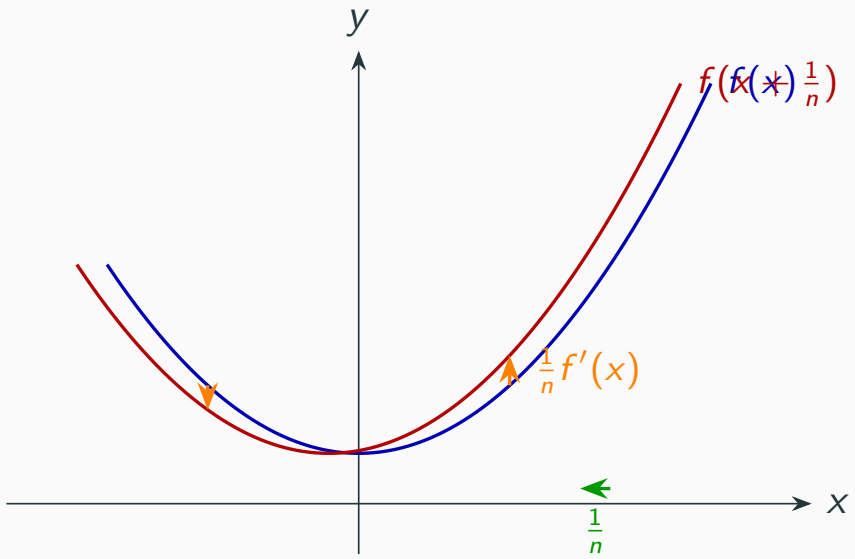

Adding $\frac{1}{n} f'(x)$ shifts the graph **up** by a slope-dependent amount — which is the same as shifting **left** by $\frac{1}{n}$.

The operator $I + \frac{1}{n}\frac{d}{dx}$ is approximately **"shift left by $\frac{1}{n}$"**.

Apply n Times

Each application shifts by $\frac{1}{n}$. Apply n times:

$$f(x) \longrightarrow f\left(x + \frac{1}{n}\right) \longrightarrow f\left(x + \frac{2}{n}\right) \longrightarrow \cdots \longrightarrow f(x+1)$$

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The Shift Operator

$$\boxed{\exp\left(\frac{d}{dx}\right)[f](x) = f(x+1)}$$

The exponential of the derivative **is** the shift-by-one operator!

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The exponential of the derivative **is** the shift-by-one operator!

More generally: $\exp\left(s\,\frac{d}{dx}\right)[f](x) = f(x+s)$ for any s.

Taylor's Theorem IS the Binomial Expansion

Now use the power series for the exponential:

$$e^{d/dx} = I + \frac{d}{dx} + \frac{1}{2!}\frac{d^2}{dx^2} + \frac{1}{3!}\frac{d^3}{dx^3} + \cdots$$

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Apply to f:

$$\underbrace{e^{d/dx}[f](x)}_{=\,f(x+1)} = f(x) + f'(x) + \frac{f''(x)}{2!} + \frac{f'''(x)}{3!} + \cdots$$

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Taylor's Theorem

$$\boxed{f(x+s) = f(x) + s\,f'(x) + \frac{s^2}{2!}\,f''(x) + \frac{s^3}{3!}\,f'''(x) + \cdots}$$

Taylor expansion is nothing more than the binomial expansion applied to the derivative operator.

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Exercise

Exercise. Using the operator $\exp\left(s\,\frac{d}{dx}\right)$, deduce the formula for $f(x+s)$ for general s.

Hint: Replace $\frac{d}{dx}$ by $s\frac{d}{dx}$ in the limit definition:

$$\exp\left(s\,\frac{d}{dx}\right) = \lim_{n\to\infty}\left(I + \frac{s}{n}\frac{d}{dx}\right)^n$$

Each application shifts by $\frac{s}{n}$. After n applications: total shift $= s$.

Matrix Exponential

Systems of Differential Equations

What if the speed of **each** quantity depends on **all** quantities?

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Written out:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\begin{aligned} y_1' &= -y_2 \\ y_2' &= 2y_1 + 3y_2 \end{aligned}$$

The speeds are **coupled** — we cannot solve them separately.

Same Discrete Trick, Matrix Version

Tiny time step $\frac{1}{n}$:

$$\mathbf{y}\left(\frac{1}{n}\right) \approx \mathbf{y}(0) + \frac{1}{n} A\,\mathbf{y}(0) = \left(I + \frac{A}{n}\right)\mathbf{y}(0)$$

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After ns steps (reaching time s):

$$\mathbf{y}(s) = \left(I + \frac{A}{n}\right)^{ns} \mathbf{y}(0)$$

Definition of Matrix Exponential

As $n \to \infty$:

Definition

$$e^A := \lim_{n\to\infty} \left(I + \frac{A}{n} \right)^n$$

Same definition as before — replace the number x with a matrix A, and 1 with I.

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But how do we compute e^{As}?

The Spectral Formula — The Payoff

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The key observation: $\left(I + \frac{A}{n}\right)^{ns}$ is a **polynomial** in A (for each finite n).

So by the spectral formula:

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As $n \to \infty$: each $\left(1 + \frac{\lambda_i}{n}\right)^{ns} \to e^{\lambda_i s}$.

Matrix Exponential via Spectral Decomposition

$$\boxed{e^{As} = e^{\lambda_1 s}\, P_1 + e^{\lambda_2 s}\, P_2}$$

Concrete Calculation

$A = \begin{pmatrix} 0 & -1 \\ 2 & 3 \end{pmatrix}$, $\det(tI - A) = (t-1)(t-2)$. Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$.

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Spectral projections (from L15):

$$P_1 = \frac{A - 2I}{1-2} = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}, \qquad P_2 = \frac{A - I}{2-1} = \begin{pmatrix} -1 & -1 \\ 2 & 2 \end{pmatrix}$$

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Solution:

$$\begin{pmatrix} y_1(s) \\ y_2(s) \end{pmatrix} = e^{As} \begin{pmatrix} y_1(0) \\ y_2(0) \end{pmatrix} = \begin{pmatrix} (2e^s - e^{2s})\, y_1(0) + (e^s - e^{2s})\, y_2(0) \\ (-2e^s + 2e^{2s})\, y_1(0) + (-e^s + 2e^{2s})\, y_2(0) \end{pmatrix}$$

Geometric Understanding via Eigenspaces

Decompose the initial condition using the spectral projections:

$$\mathbf{y}(0) = P_1\,\mathbf{y}(0) + P_2\,\mathbf{y}(0) = \mathbf{w}(0) + \mathbf{u}(0)$$

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Total:

$$\mathbf{y}(s) = e^{s}\,\mathbf{w}(0) + e^{2s}\,\mathbf{u}(0)$$

Each eigenspace component grows at its own rate. The **larger eigenvalue dominates** as $s \to \infty$.

Complex Eigenvalues

What If Eigenvalues Are Complex?

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Using Euler's formula:

- $e^{i\theta} = \cos\theta + i \sin\theta$
- $e^{-i\theta} = \cos\theta - i \sin\theta$

Computing the Projections

Eigenvalues: $\lambda_1 = i$, $\lambda_2 = -i$.

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Similarly: $P_2 = \dfrac{R - iI}{-2i} = \dfrac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$

Note: $P_2 = \overline{P_1}$ — conjugate pair of projections for conjugate pair of eigenvalues (the symmetry principle from L17!).

The Result: Rotation by Any Angle

$$e^{R\theta} = e^{i\theta} \cdot \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} + e^{-i\theta} \cdot \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

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Look at the $(1,1)$ entry: $\frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \cos\theta$

Look at the $(2,1)$ entry: $\frac{1}{2}(-i\, e^{i\theta} + i\, e^{-i\theta}) = \frac{i(e^{-i\theta} - e^{i\theta})}{2} = \sin\theta$

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Collecting all entries:

Rotation Matrix

$$\boxed{e^{R\theta} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}$$

The matrix exponential of the 90° rotation generator gives **rotation by any angle θ**.

The Big Picture: Real vs. Complex Eigenvalues

Eigenvalue	$e^{\lambda s}$	**Meaning**
$\lambda > 0$	grows	Exponential **growth**
$\lambda < 0$	decays	Exponential **decay**
$\lambda = i\omega$	rotates	Pure **oscillation**
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Real eigenvalues produce growth or decay.

Complex eigenvalues produce rotation and oscillation.

The spectral formula $e^{As} = \sum e^{\lambda_i s} P_i$ handles **all cases uniformly**.

Summary

One Definition, Four Faces

The Definition

$$e^X = \lim_{n\to\infty}\left(I + \frac{X}{n}\right)^n$$

X	e^X	Meaning
$x \in \mathbb{R}$	e^x	Growth / decay
$i\theta$	$\cos\theta + i\sin\theta$	Rotation (Euler)
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$\frac{d}{dx}$	Shift operator	Taylor's theorem

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Repeated tiny nudges compound into a smooth transformation.

The binomial expansion of the limit gives the power series. Everything is one idea.